# NORMALIZATION COMMUTES WITH HENSELIZATION AND STRICT HENSELIZATION AT A POINT

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 5438-5450 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5452-5496, the complete author proof (43 body lines).

*Editorial note.* Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Normalization of a locally Noetherian scheme, integral closure, and the description of henselization and strict henselization as filtered colimits of etale algebras; the reductions are retained.

*Source reference.* Original lemma algebra-lemma-flat-going-down: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5496–5496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma algebra-lemma-integral-closure-commutes-colim-smooth: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5496–5496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma algebra-lemma-integral-closure-commutes-etale: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5496–5496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma algebra-lemma-minimal-prime-image-minimal-prime: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5496–5496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma more-algebra-lemma-fibres-henselization: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5496–5496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma more-algebra-lemma-henselization-noetherian: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5496–5496.

*Date*: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma morphisms-definition-normalization: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5496–5496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma morphisms-lemma-description-normalization: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5496–5496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma morphisms-lemma-integral-closure: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5496–5496. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1** (Normalization and henselization)**.** *Let $X$ be a locally Noetherian scheme. Let $\nu : X^\nu \to X$ be the normalization morphism. Then for any point $x \in X$ the base change*

$$X^\nu \times_X \mathrm{Spec}(\mathcal{O}^h_{X,x}) \to \mathrm{Spec}(\mathcal{O}^h_{X,x}), \quad \textit{resp.} \quad X^\nu \times_X \mathrm{Spec}(\mathcal{O}^{sh}_{X,x}) \to \mathrm{Spec}(\mathcal{O}^{sh}_{X,x})$$

*is the normalization of* $\mathrm{Spec}(\mathcal{O}^h_{X,x})$*, resp.* $\mathrm{Spec}(\mathcal{O}^{sh}_{X,x})$*.*

*Proof.* Let $\eta_1, \ldots, \eta_r$ be the generic points of the irreducible components of $X$ passing through $x$. The base change of the normalization to $\mathrm{Spec}(\mathcal{O}_{X,x})$ is the spectrum of the integral closure of $\mathcal{O}_{X,x}$ in $\prod \kappa(\eta_i)$. This follows from our construction of the normalization of $X$ in Morphisms, Definition morphisms-definition-normalization and Morphisms, Lemma morphisms-lemma-integral-closure; you can also use the description of the normalization in Morphisms, Lemma morphisms-lemma-description-normalization. Thus we reduce to the following algebra problem. Let $A$ be a Noetherian local ring; recall that this implies the henselization $A^h$ and strict henselization $A^{sh}$ are Noetherian too (More on Algebra, Lemma more-algebra-lemma-henselization-noetherian). Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ be its minimal primes. Let $A'$ be the integral closure of $A$ in $\prod \kappa(\mathfrak{p}_i)$. Problem: show that $A' \otimes_A A^h$, resp. $A' \otimes_A A^{sh}$ is constructed from the Noetherian local ring $A^h$, resp. $A^{sh}$ in the same manner.

Since $A^h$, resp. $A^{sh}$ are colimits of étale $A$-algebras, we see that the minimal primes of $A$ and $A^{sh}$ are exactly the primes of $A^h$, resp. $A^{sh}$ lying over the minimal primes of $A$ (by going down, see Algebra, Lemmas algebra-lemma-flat-going-down and algebra-lemma-minimal-prime-image-minimal-prime). Thus More on Algebra, Lemma more-algebra-lemma-fibres-henselization tells us that $A^h \otimes_A \prod \kappa(\mathfrak{p}_i)$, resp. $A^{sh} \otimes_A \prod \kappa(\mathfrak{p}_i)$ is the product of the residue fields at the minimal primes of $A^h$, resp. $A^{sh}$. We know that taking the integral closure in an overring commutes with étale base change, see Algebra, Lemma algebra-lemma-integral-closure-commutes-etale. Writing $A^h$ and $A^{sh}$ as a limit of étale $A$-algebras we see that the same thing is true for the base change to $A^h$ and $A^{sh}$ (you can also use the more general Algebra, Lemma algebra-lemma-integral-closure-commutes-colim-smooth). □